NANDINI WALIA

Student

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EXPERIENCE

Avionics and Coding Engineer

Team Albatross

08/2024 Present | Vellore, India

- Designed and tested propulsion systems by evaluating thrust-to-weight ratios, current draw, and thermal limits
- Integrated sensors to develop a telemetry system using ESP32 to acquire real-time flight parameters including altitude, GPS position, temperature and acceleration.
- Interfaced multiple sensors using UART and I2C protocols
- Modeled aircraft dynamics and control loops using MATLAB Simulink to simulate flight behaviour
- Contributed to multiple award-winning projects, including world-ranked aircraft systems

EDUCATION

Bachelor of Technology in Electrical and Electronics Engineering

Vellore Institute of Technology

08/2023 Present | Vellore

- Coursework: DC Machines and Transformers, Power Systems Engineering, Power Electronics, Analog Electronics, Digital Electronics, Control Systems, Microprocessors and Microcontrollers, Signals and Systems

SKILSS

Equipment

Oscilloscope, Vector Network Analyzer, Function Generator, Digital Multimeter, DC Machines, Transformers, PID Controllers, Servo motors.

Electronics

ESP32, ESP32 S3, STM32, Raspberry Pi, Arduino, Op-Amps, BJTs

Programming

C , Python, Java

Software Tools

MATLAB, Simulink, OrCAD PSpice, Altium, KiCAD, SolidWorks, Ecalc, Scikit-learn

PROJECTS

Propulsion System for RC aircrafts

- Manually iterated 100's of combination of ESC-motor-propeller to achieve 80% thrust efficiency
- Designing of a real-time telemetry system using ESP32 MCU and sensors including BMP280, NEO 6M, MPU6050, and DS18B20
- Integrated sensor data via I2C and UART interfaces to enable live ground-station feedback and analysis
- Contributed to award-winning aircraft models in SAE Aero Design West 2025 and SAEISS DDC2025.

On-Board Charger (iOBC for Two-Wheeler EVs

- Designed a high-efficiency on-board charger using a boost converter and phase-shifted full-bridge topology for PFC and bidirectional charging V2G, V2L
- Achieved 95% efficiency and 0.99 power factor through Zero Voltage Switching
- Integrated wide-bandgap devices, modular design, and ensured IEC61851/UL2202 safety compliance
- Included control simulations, performance graphs, and ML-based EV adoption analysis.

Real-Time Telemetry System

- Designed a real-time telemetry system using ESP32 microcontrollers and sensors including BMP280, NEO 6M, MPU6050, and DS18B20
- Integrated sensor data via I2C and UART interfaces to enable live ground-station feedback and analysis
- Designed a Simulink model to simulate and predict aircraft behavior, supporting pre-flight validation and control tuning
- Conducted a comparative analysis of the collected data versus simulation outcomes for documentation purposes

INTERESTS

Computational Modeling of Electronic Components

Interested in using computational methods to model and simulate electronic components significant for developing advanced power electronic systems

Machine Learning in Power Electronics

I am passionate about applying machine learning to enhance performance of power electronic systems. I am interested in developing models that could predict efficiency, detect faults early, and optimize system parameters

Aerodynamics

Interest in aviation made me join Team Albatross involving the design of RC aircarfts. My work in the team inclined towards the working of avionic components and how they influence the aerodynamic performance of the aircarft.